

AWS Snow Family

The **AWS Snow Family** is a group of physical devices designed by Amazon Web Services to help customers **move large amounts of data in and out of AWS** when online data transfer is **impractical due to bandwidth, time, or cost constraints**.

◆ Overview of AWS Snow Family

Device	Purpose	Storage Capacity	Key Use Cases
Snowcone	Small-scale data transfer and edge computing	8 TB usable	Edge computing, disconnected or limited network areas
Snowball Edge	Large-scale data migration and edge computing	80 TB to 210 TB usable	Data migration, edge analytics, ML inference, local compute
Snowmobile	Exabyte-scale data transfer	Up to 100 PB per truck	Massive data center migrations

◆ Why Use the AWS Snow Family?

- ✓ Limited or no internet access
 - ✓ Need to move PB-scale data faster than online transfer
 - ✓ Cost-effective for large datasets
 - ✓ Offers local compute and storage for edge use cases
 - ✓ Data is encrypted and secure end-to-end
-

◆ Components of the AWS Snow Family

1. AWS Snowcone

- **Smallest device** in the Snow family

- **Rugged, portable**, and can run on battery or DC power
- Ideal for **edge locations, IoT, or rugged environments**
- Supports **AWS IoT Greengrass** and **EC2-compatible compute**
- Storage: **8 TB usable HDD or SSD**
- **Offline and online** data transfer (with AWS DataSync agent)

Use Cases:

- Edge computing in disconnected locations
 - Disaster zones, military, ships, etc.
 - Drone or vehicle-mounted data collection
-

2. AWS Snowball Edge

- Medium-sized, **rugged** devices with two variants:
 - **Snowball Edge Storage Optimized** (210 TB usable storage)
 - **Snowball Edge Compute Optimized** (42 TB usable storage + GPU)
- Can run **EC2 instances, Lambda functions**, and **S3-compatible storage**
- Supports clustering multiple devices
- Supports **NFS, SMB**, and **HDFS** for local access

Use Cases:

- Local data processing at the edge
- Machine learning inference at remote sites
- Mass data migration (PB scale)

- Temporary storage for applications
-

3. AWS Snowmobile

- A **shipping container-sized** data transfer solution (45-foot truck)
- Up to **100 PB** of data per Snowmobile
- Uses **tamper-proof, temperature-controlled, and GPS-tracked** truck
- Requires AWS personnel for on-site setup and security

Use Cases:

- Entire data center migrations
 - Archiving massive datasets
 - Regulatory/compliance-driven offline migrations
-



Security Features

- **End-to-end encryption** using 256-bit keys (managed by AWS KMS)
 - Devices are **tamper-resistant and tamper-evident**
 - Data is encrypted **before it leaves your premises**
 - Automatically **erased** after successful data import
-



Workflow for Using Snow Devices

1. **Create a job** in the AWS Management Console

2. AWS **prepares and ships** the device to your location
 3. Connect device and **transfer data** (via S3, NFS, HDFS, etc.)
 4. **Ship device back** to AWS
 5. AWS **ingests data into your S3 bucket**
 6. Device is securely **wiped and reused**
-



Notes for Exams/Interviews

- **Snowcone** = lightweight, edge use
 - **Snowball** = mid-range, compute + storage
 - **Snowmobile** = massive data center migration
 - All devices support **secure, offline transfer**
 - Snowball and Snowcone support **local compute**
 - **DataSync** can be used with Snowcone for online transfers
 - Devices are ordered and managed via **AWS Console**
 - Often used when **upload over the internet is too slow**
-



Related AWS Services

- **AWS DataSync** – Automates data movement to and from AWS
- **AWS Transfer Family** – For SFTP/FTP/FTPS online transfers
- **AWS IoT Greengrass** – For running local compute on edge devices

Here's a detailed update on **AWS Snow Family pricing and related considerations**:

Pricing Breakdown

AWS Snowcone

- **Service Fee per Job:**
 - \$60 for standard Snowcone,
 - \$150 for Snowcone SSD. ([Amazon Web Services](#))
- **Included Usage:** 5 days of device usage (shipping days not counted). ([Amazon Web Services](#))
- **Additional Per-Day Fee:**
 - \$6/day for Snowcone,
 - \$15/day for SSD version. ([Amazon Web Services](#))
- **Monthly Rental Option:**
 - \$150/month for standard,
 - \$400/month for SSD—no extra shipping or daily fees. ([Amazon Web Services](#))
- **Data Transfer:**
 - Inbound to S3 is free.
 - Outbound from S3 varies by region (e.g. \$0.04/GB in Mumbai). ([Amazon Web Services](#))

AWS Snowball Edge

- **On-Demand Pricing:**

- Includes a per-job service fee, per-day charges after a free period, plus shipping. Pricing varies by device type. ([Amazon Web Services](#), bluexp.netapp.com)
- **Long-Term / Commitment-Based Pricing:**
 - Options for monthly, 1-year, or 3-year upfront commitments. Up to 62% savings compared to on-demand. ([Amazon Web Services](#), bluexp.netapp.com, [AWS Documentation](#))
- **Sample US East Pricing (Storage-Optimized):**
 - On-demand: approx \$300 per job, \$30/day after initial free days. (bluexp.netapp.com)
 - 1-year commitment: ~\$15,330 (\$42/day equivalent). (bluexp.netapp.com)
- **Loss/Damage Fees:**
 - Storage-Optimized: \$12,500
 - Compute-Optimized: \$20,000
 - Compute-Optimized with GPU: \$30,000 (bluexp.netapp.com)

AWS Snowmobile

- **Pricing Model:** \$0.005 per GB per month, based on provisioned capacity—not actual usage. ([Logicata - AWS Managed Services](#))
- **Additional Costs:** Standard S3 storage and request fees apply. ([Logicata - AWS Managed Services](#))
- **Note:** AWS Snowmobile is typically ordered via the **Sales team**, not the Console. ([Logicata - AWS Managed Services](#))

Additional Insights & Real-World Takeaways

- **Availability & Discontinuation**

- As of November 2024, earlier-generation Snowball and Snowcone devices were flagged for discontinuation, though support continues through November 2025. ([Amazon Web Services](#))
- **Long-Term Pricing Benefits**
 - Committing for 1 or 3 years offers significant cost savings on Snowball Edge workloads. ([Amazon Web Services](#), [AWS Documentation](#))
- **Shipping Costs**
 - Shipping can be surprisingly high. One user in Australia reported ~236 AUD plus 60 AUD for fees—over 3x the base device cost. ([Reddit](#))
- **User-Reported Loss Fees**
 - Example: Losing a Snowball Edge Storage-Optimized could cost \$12,500 (confirming AWS documentation). ([Reddit](#))
- **Service Delays**
 - Some customers have experienced long delays in device fulfillment—up to 2+ months. ([Reddit](#))

Summary Table

Device	Pricing Highlights
Snowcone	\$60 job fee + \$6/day (std) or \$150 job + \$15/day (SSD); or \$150/\$400 monthly rental
Snowball Edge (On-Demand)	~ \$300/job + ~\$30/day (Storage-Optimized); higher for compute/GPU options
Snowball Edge (Commitment)	1-year savings (~\$15k) or 3-year deep discounts—no daily or shipping fees
Snowmobile	\$0.005/GB/month on provisioned capacity, via Sales only
Additional Fees	S3 storage/API costs, data egress, shipping, and potential loss/damage penalties
